MINI PROJECT ASSIGNMENT

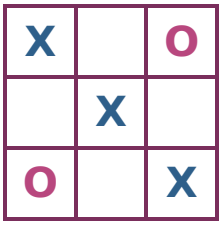
Tic-Tac-Toe Game

Using Python (2-Player, Console Based)

DIFFICULTY	LANGUAGE	INTERFACE
★☆☆☆☆ Beginner	**Python 3.x**	**Command Line (CLI)**

Objective

Design and build a simple two-player Tic-Tac-Toe game that runs in the console. The project should demonstrate your understanding of core Python fundamentals such as functions, loops, conditional statements, lists, and basic input validation — without requiring any external libraries or a database.



Game Rules

- The game is played on a 3×3 grid by two players, taking turns.
- Player 1 uses **X** and Player 2 uses **O**.
- Players alternate placing their mark in an empty cell.
- The first player to get three of their marks in a row, column, or diagonal wins.
- If all 9 cells are filled and no one has won, the game ends in a draw.

Project Requirements

1 Game Board

- Represent the 3×3 board using a suitable data structure (e.g. a list of lists, or a single list of 9 elements).
- Display the board neatly in the console after every move, with clear row and column separators.

2 Two-Player Turns

- Support two players taking turns on the same keyboard.
- Clearly display whose turn it is (Player 1 – X, or Player 2 – O) before each move.

3 Taking a Move

- Ask the current player to enter a position on the board (e.g. row and column, or a cell number 1–9).
- Place the player's mark on the board if the chosen cell is valid and empty.

4 Input Validation

The program should prevent and gracefully handle:

- Non-numeric or out-of-range input
- Selecting a cell that is already occupied
- Empty input

On invalid input, show a friendly message and ask the same player to try again, without crashing.

5 Win Detection

After every move, check whether the current player has won by completing:

- Any of the 3 rows
- Any of the 3 columns
- Either of the 2 diagonals

If a player wins, display the final board and announce the winner, then end the game.

6 Draw Detection

- If the board is completely filled and no player has won, display the final board and announce the game as a draw.

7 Play Again Option

- After a game ends (win or draw), ask the players if they would like to play another round.
- If yes, reset the board and start a new game; if no, exit with a friendly closing message.

Technical Requirements

Variables & Data Types	Functions	Conditional Statements
Loops (for / while)	Lists / 2D Lists	Input Validation
Basic Exception Handling	String Formatting	OOP (Optional)

Bonus Challenges (Optional)

Add a single-player mode vs. a simple computer opponent	Keep a score tally across multiple rounds	Let players choose their own names and symbols

Submission Requirements

Students must submit:

- Complete Python source code (.py file)
- Short README explaining how to run the game
- Well-commented and properly formatted code

Mandatory Checklist

✓ 3×3 board display	✓ Two-player turn handling
✓ Move input & validation	✓ Win detection (rows/cols/diagonals)
✓ Draw detection	✓ Play again option

Evaluation Rubric (50 Marks)

Criteria	Marks
Board Display & Game Logic	**15**
Turn Handling & Move Input	**10**
Win & Draw Detection	**10**
Input Validation & Error Handling	**8**
Code Quality & Documentation	**7**
Total	**50**